

# Appendix A

## APPENDIX A – Parameters and Performance Metrics of the IDF Equation Models

This appendix presents the detailed results of the Intensity-Duration-Frequency (IDF) equations fitting for eleven municipalities in the state of Goiás. The following tables detail the fitted parameters and the statistical performance metrics — Coefficient of Determination ( $R^2$ ), Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Nash-Sutcliffe Efficiency (NSE) — for each studied location.

The results are divided based on the two methodological approaches tested: the **One Step Procedure (OSP)**, characterized by the simultaneous fitting of all durations aiming at a single equation, and the **Standard Procedure (SP)**. For each method, the **GEV** (Generalized Extreme Value) and **Gumbel** probability distributions were evaluated.

It should be noted that, for the proper use of the proposed equations, the duration  $D$  must be input in hours to yield the intensity  $I$  in  $mm/h$ .

## A.1 Results for the One Step Procedure (OSP)

Table 1: Parameters and performance metrics of the OSP-GEV model (One Step Procedure and GEV distribution)

| Municipality             | Station<br>Code | $R^2$ | RMSE | MAE  | NSE  | $a'$  | $b'$ | $c$  | $d$  | $k$  |
|--------------------------|-----------------|-------|------|------|------|-------|------|------|------|------|
| ÁGUAS LINDAS<br>DE GOIÁS | 520025803A      | 0.99  | 2.87 | 1.40 | 0.99 | 53.25 | 0.00 | 0.33 | 0.89 | 0.15 |
| ALEXÂNIA                 | 520030801A      | 0.98  | 3.95 | 1.84 | 0.98 | 55.82 | 0.00 | 0.39 | 0.93 | 0.20 |
| ANÁPOLIS                 | 520110801A      | 0.99  | 3.19 | 1.56 | 0.99 | 78.71 | 0.00 | 0.46 | 1.00 | 0.11 |
| APARECIDA<br>DE GOIÂNIA  | 520140506A      | 0.99  | 2.91 | 1.30 | 0.99 | 61.36 | 0.00 | 0.46 | 0.98 | 0.16 |
| CALDAS<br>NOVAS          | 520450801A      | 0.98  | 3.75 | 1.94 | 0.97 | 53.76 | 0.00 | 0.54 | 0.98 | 0.21 |
| CATALÃO                  | 520510901A      | 0.92  | 8.74 | 3.05 | 0.92 | 76.42 | 0.00 | 0.60 | 1.00 | 0.19 |
| ITUMBIARA                | 521150301A      | 0.99  | 2.22 | 1.21 | 0.99 | 69.35 | 0.01 | 0.56 | 0.90 | 0.13 |
| JATAÍ                    | 521190901A      | 1.00  | 1.58 | 0.95 | 1.00 | 72.76 | 0.00 | 0.53 | 0.99 | 0.10 |
| NIQUELÂNDIA              | 521460602A      | 0.99  | 2.70 | 1.48 | 0.99 | 62.30 | 0.00 | 0.37 | 0.93 | 0.15 |
| URUAÇU                   | 522160101A      | 0.99  | 2.31 | 1.10 | 0.99 | 40.36 | 0.00 | 0.37 | 0.93 | 0.23 |
| URUANA                   | 522170001A      | 1.00  | 1.63 | 0.96 | 1.00 | 45.71 | 0.00 | 0.33 | 0.84 | 0.14 |

Table 2: Parameters and performance metrics of the OSP-Gumbel model (One Step Procedure and Gumbel distribution)

| Municipality             | Station<br>Code | $R^2$ | RMSE | MAE  | NSE  | $a'$  | $b'$ | $c$  | $d$  |
|--------------------------|-----------------|-------|------|------|------|-------|------|------|------|
| ÁGUAS LINDAS<br>DE GOIÁS | 520025803A      | 0.99  | 2.64 | 1.24 | 0.99 | 9.48  | 5.57 | 0.32 | 0.89 |
| ALEXÂNIA                 | 520030801A      | 0.99  | 3.51 | 1.73 | 0.99 | 14.98 | 3.87 | 0.43 | 0.95 |
| ANÁPOLIS                 | 520110801A      | 0.99  | 2.98 | 1.51 | 0.99 | 10.40 | 7.59 | 0.47 | 1.00 |
| APARECIDA DE<br>GOIÂNIA  | 520140506A      | 0.99  | 2.62 | 1.17 | 0.99 | 12.93 | 4.78 | 0.48 | 0.99 |
| CALDAS NOVAS             | 520450801A      | 0.98  | 3.47 | 1.73 | 0.98 | 15.23 | 3.23 | 0.50 | 0.95 |
| CATALÃO                  | 520510901A      | 0.93  | 8.21 | 2.96 | 0.93 | 18.20 | 4.14 | 0.57 | 1.00 |
| ITUMBIARA                | 521150301A      | 0.99  | 2.25 | 1.20 | 0.99 | 10.16 | 6.91 | 0.56 | 0.90 |
| JATAÍ                    | 521190901A      | 1.00  | 1.48 | 0.91 | 1.00 | 7.72  | 9.46 | 0.53 | 0.99 |
| NIQUELÂNDIA              | 521460602A      | 0.99  | 2.30 | 1.27 | 0.99 | 10.79 | 5.78 | 0.37 | 0.93 |
| URUAÇU                   | 522160101A      | 0.99  | 1.79 | 0.89 | 0.99 | 12.49 | 3.34 | 0.41 | 0.94 |
| URUANA                   | 522170001A      | 1.00  | 1.54 | 0.95 | 1.00 | 7.27  | 6.45 | 0.35 | 0.84 |

## A.2 Results for the Standard Procedure (SP)

Table 3: Parameters and performance metrics of the SP-GEV model (Standard Procedure and GEV distribution)

| Municipality          | Station Code | $R^2$ | RMSE  | MAE  | NSE  | $a$    | $b$  | $c$  | $d$  |
|-----------------------|--------------|-------|-------|------|------|--------|------|------|------|
| ÁGUAS LINDAS DE GOIÁS | 520025803A   | 0.97  | 5.32  | 2.58 | 0.96 | 54.91  | 0.08 | 0.27 | 0.86 |
| ALEXÂNIA              | 520030801A   | 0.95  | 7.69  | 3.99 | 0.93 | 81.37  | 0.10 | 0.63 | 1.00 |
| ANÁPOLIS              | 520110801A   | 0.97  | 8.04  | 4.61 | 0.94 | 61.84  | 0.17 | 0.35 | 1.15 |
| APARECIDA DE GOIÂNIA  | 520140506A   | 0.97  | 5.30  | 2.32 | 0.96 | 63.14  | 0.11 | 0.45 | 1.00 |
| CALDAS NOVAS          | 520450801A   | 0.93  | 7.01  | 4.25 | 0.91 | 84.89  | 0.08 | 0.90 | 1.00 |
| CATALÃO               | 520510901A   | 0.86  | 13.09 | 6.77 | 0.82 | 120.78 | 0.09 | 1.02 | 1.00 |
| ITUMBIARA             | 521150301A   | 0.95  | 6.43  | 3.47 | 0.94 | 34.15  | 0.17 | 0.01 | 0.62 |
| JATAÍ                 | 521190901A   | 0.99  | 3.49  | 2.79 | 0.99 | 59.04  | 0.06 | 0.29 | 0.78 |
| NIQUELÂNDIA           | 521460602A   | 0.98  | 5.57  | 2.84 | 0.97 | 65.14  | 0.06 | 0.32 | 0.89 |
| URUAÇU                | 522160101A   | 0.96  | 4.61  | 2.28 | 0.96 | 50.26  | 0.17 | 0.52 | 1.00 |
| URUANA                | 522170001A   | 0.99  | 3.52  | 2.16 | 0.98 | 40.99  | 0.15 | 0.28 | 0.93 |

Table 4: Parameters and performance metrics of the SP-Gumbel model (Standard Procedure and Gumbel distribution)

| Municipality          | Station Code | $R^2$ | RMSE  | MAE  | NSE  | $a$   | $b$  | $c$  | $d$  |
|-----------------------|--------------|-------|-------|------|------|-------|------|------|------|
| ÁGUAS LINDAS DE GOIÁS | 520025803A   | 0.98  | 4.02  | 2.01 | 0.98 | 67.29 | 0.14 | 0.53 | 1.00 |
| ALEXÂNIA              | 520030801A   | 0.97  | 6.26  | 3.16 | 0.95 | 73.85 | 0.18 | 0.62 | 1.00 |
| ANÁPOLIS              | 520110801A   | 0.99  | 3.96  | 1.94 | 0.99 | 75.89 | 0.11 | 0.45 | 1.00 |
| APARECIDA DE GOIÂNIA  | 520140506A   | 0.98  | 4.36  | 2.20 | 0.98 | 69.50 | 0.15 | 0.58 | 1.00 |
| CALDAS NOVAS          | 520450801A   | 0.96  | 5.60  | 3.45 | 0.94 | 76.02 | 0.18 | 0.92 | 1.00 |
| CATALÃO               | 520510901A   | 0.91  | 12.71 | 5.80 | 0.83 | 85.13 | 0.25 | 0.70 | 1.00 |
| ITUMBIARA             | 521150301A   | 0.99  | 3.41  | 2.08 | 0.98 | 90.16 | 0.11 | 0.85 | 0.98 |
| JATAÍ                 | 521190901A   | 0.99  | 2.37  | 1.81 | 0.99 | 65.20 | 0.09 | 0.43 | 0.88 |
| NIQUELÂNDIA           | 521460602A   | 0.99  | 3.82  | 1.99 | 0.99 | 63.02 | 0.13 | 0.38 | 0.93 |
| URUAÇU                | 522160101A   | 0.97  | 4.60  | 2.55 | 0.96 | 48.87 | 0.18 | 0.50 | 0.95 |
| URUANA                | 522170001A   | 0.99  | 2.61  | 1.46 | 0.99 | 45.39 | 0.12 | 0.31 | 0.83 |